CARMELLOSE (ARTIFICIAL TEARS)

Dry eye
Preservative-free preferred

CHEMICAL NAME Carmellose sodium (carboxymethylcellulose sodium) 0.5–1% in buffered solution
CLASS Ocular lubricant (tear substitute)
TRADE NAMES Celluvisc · Refresh Tears · Optive · Cellufresh

THERAPEUTIC USES / INDICATIONS

- Dry eye disease (keratoconjunctivitis sicca), Sjögren's syndrome.
- Eye protection in unconscious/sedated patients (ICU) and incomplete eyelid closure (Bell's palsy).
- Eye irritation from screens, contact lenses, environment.

PHARMACODYNAMICS

- Mucoadhesive polymer lubricates and protects the corneal surface, stabilises tear film, reduces friction and irritation.

ADMINISTRATION

- **Eye drops** 1–2 drops as needed; single-dose units (preservative-free) for frequent use or contact lens wearers.
- Avoid touching tip to eye.
- Wait 5–10 min after other eye drops.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Transient blurred vision (thicker formulations).
- Mild stinging; crusting on lashes.

ADVERSE EFFECTS (SERIOUS)

- Allergic reactions (rare); preservative toxicity with frequent use of preserved products.

DOSAGES

- 0.5–1%: 1–2 drops in each affected eye as often as needed (e.g. 4–6 times daily or hourly).
- ICU: per eye-care protocol (with lid closure/taping or ointment).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to components.

KEY INTERACTIONS

- May dilute other eye drops — separate by 5–10 min.

PHARMACOKINETICS

- Not absorbed; viscous polymer remains on ocular surface and drains via tear ducts.

NURSING MANAGEMENT

- ICU: eye care every 2–4 h; assess for exposure keratopathy (corneal damage); ensure lid closure.
- Assess dry-eye symptoms, blink rate.
- Teach technique.

PATIENT EDUCATION

- Use as often as needed; preservative-free single-dose vials for frequent use.
- Take regular screen breaks, blink fully, use humidifiers.
- See a doctor if pain, redness or reduced vision occurs.

KEY POINTS

- Artificial tears; lubricate cornea.
- Essential ICU eye care.
- Use as often as needed.
- Preservative-free for frequent use.